Semantic and spatial similarity in geographic information retrieval

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ABSTRACT ORIGINAL

People use the similarity to store and retrieve information, also for learning and concept formation. Similarity plays a fundamental role in many applications such as decision-making systems, data mining and pattern recognition. The same applies to the spatial similarity in the processes of recovery and integration of spatial information. In this paper a methodology based on the semantic processing of geospatial information is proposed. It consists of five stages: conceptualization, synthesis, application processing, retrieval and management. It uses ontology of New Genetic Soil Classification of Cuba and applying the measure of semantic similarity DISC combined with an implementation of model TDD (Topology-Direction-Distance) to restore and support the user in the selection of spatial scenes. As a case study is considered the town of San José de Las Lajas located in the province of Mayabeque in western of Cuba.